# Financing the Air We Breathe: Climate vs. Non-Climate Aid and Local CO2 Emissions

## Staggered Event-Study and Dose–Response Evidence from Global ADM2 Data


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**Abstract**

This paper asks whether climate aid decarbonizes where it lands, and how its effects compare to otherwise similar development finance. I assemble a global ADM2–year panel (2000–2022) linking geocoded projects to EDGAR CO2 and estimate dynamic effects with a staggered DiD that uses not-yet-treated controls, complemented by a post-treatment dose–response design. Two results emerge. First, climate portfolios lead to at most zero and on average modest declines in local log-CO2, which grow more negative as cumulative climate exposure increases. Second, non-climate aid raises local emissions in the medium run, consistent with development-driven scale effects. Identification is supported by flat pre-trends, placebo event-times, and robustness to excluding China. To probe mechanisms, I use VIIRS night-time lights (NTL) as a development proxy: non-climate projects raise NTL in mid-brightness ADM2 (more activity), while in already bright places NTL responses are muted—consistent with LED/dimming and land-use substitution toward transport/industrial surfaces—yet CO still rises. Together, the evidence indicates: climate aid decarbonizes weakly to moderately; non-climate aid increases emissions by expanding local economic activity, with NTL patterns tracking that expansion where radiance remains informative.



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# Introduction

International and domestic climate finance increasingly flows through, or must be implemented by, sub-national governments. Cities and regions control large shares of infrastructure investment, land-use regulation, and service delivery. Yet credible evidence on who receives climate finance, at what cost, and with what effects remains fragmented across disciplines. This article organizes that evidence articulates a tractable conceptual framework, and maps it to empirical strategies suitable for causal evaluation.

# Literature Review

The relationship between development, climate aid/finance, and CO2 emissions is complex. Climate finance and targeted aid can reduce CO2 emissions, but the effectiveness depends on the type of aid, governance, and the stage of economic development. Economic growth and financial development alone often increase emissions unless paired with green finance or strong environmental policies.

Climate finance, especially when directed toward mitigation, is associated with lower CO2 emissions, with stronger effects in countries with higher economic development and in small island states (C.-C. Lee et al. 2022; Khan, Ahmed, and Hull 2023). Green finance and green technology innovation consistently show significant negative impacts on CO2 emissions in both developed (G7, OECD) and emerging economies (BRICS) (Sharif et al. 2022; Saeed Meo and Karim 2022; Umar and Safi 2023; Sadiq et al. 2024). Climate aid for renewable energy has a slight but transitory effect on reducing CO2 emissions, with greater impact after a certain threshold. However, its effectiveness is limited if recipient countries prioritize economic over ecological efficiency (Kablan and Chouard 2022).

Regarding Official Development Assistance (ODA), it can directly and indirectly mitigate CO2 emissions, but the effect is stronger in middle-income countries and when governance is robust. In low-income countries, the impact is limited (Barkat, Alsamara, and Mimouni 2024; S. K. Lee et al. 2020; Farooq 2021; Mahalik et al. 2021). However, aid directed toward fossil fuel development increases CO2 emissions, especially in countries with less environmental regime integration (K. E. Henderson and Sommer 2022; K. E. Henderson 2019). Strong governance and environmental policies enhance the effectiveness of aid and finance in reducing emissions (Barkat, Alsamara, and Mimouni 2024; Farooq 2021; K. E. Henderson 2019).

Recent research provides clear evidence that green finance and climate aid can significantly reduce CO2 emissions at the subnational (regional or city) level, but the effectiveness varies by region, policy design, and local economic structure.

Studies using data from hundreds of Chinese cities and provinces show that green finance policies and instruments (such as green loans and subsidies) have a significant negative effect on CO2 emissions intensity at the subnational level. The effect is stronger in eastern and non-resource-based cities, while central and western or resource-dependent regions see weaker impacts (Y. Sun, Bao, and Taghizadeh-Hesary 2023; F. Wang, Cai, and Elahi 2021; Zhao et al. 2024; H. Li et al. 2025). Uniform national policies may be less effective than regionally tailored approaches. Incentive-based policies work better in developed eastern provinces, while a mix of supportive and pressure policies (like carbon taxes) are needed in less developed central regions (Y. Sun, Bao, and Taghizadeh-Hesary 2023; H. Li et al. 2025). Green finance reduces emissions by supporting green technology innovation, upgrading industrial structure, and promoting renewable energy adoption. Debt-based green finance instruments are more effective than equity-based ones in reducing emissions at the city level (F. Wang, Cai, and Elahi 2021; Zhao et al. 2024) . At the micro level, green finance helps firms reduce emissions through lower financing costs, technological innovation, and external supervision. However, emission reduction is less pronounced in heavy-polluting and resource-based industries (Zhao et al. 2024).

Green finance projects, those specifically designed to support low-carbon, environmentally sustainable activities, consistently outperform traditional development finance projects in reducing CO2 emissions at the subnational (regional or city) level. Green finance significantly reduces local and neighboring CO2 emissions, with a 1% increase in green financial development linked to a 3% decrease in regional emissions (J. Sun et al. 2022; Chen and Chen 2021; Su, Qiao, and Xu 2024). The effect is strongest in economically advanced or eastern regions and in non-resource-based cities, where financial markets and green policy implementation are more mature (Chen and Chen 2021; Su, Qiao, and Xu 2024). Green finance also creates "ripple effects," reducing emissions in adjacent areas by promoting green technology and industrial upgrades (J. Sun et al. 2022; Chen and Chen 2021; Su, Qiao, and Xu 2024).

General development finance (not specifically climate-related) can have mixed or even adverse effects on emissions. In some cases, it increases emissions by fueling economic growth and energy consumption, especially in less developed or resource-dependent regions (Chen and Chen 2021; Su, Qiao, and Xu 2024; L. Wang, Yang, and Cai 2024; Mumuni and Hamadjoda Lefe 2023) . The emission-reducing effect of traditional finance is weaker, and in some regions, it may actually exacerbate pollution unless paired with strong environmental regulations or targeted toward green sectors (Chen and Chen 2021; L. Wang, Yang, and Cai 2024). Traditional finance lacks these targeted mechanisms and may inadvertently support high-emission sectors (Chen and Chen 2021; L. Wang, Yang, and Cai 2024).

At the subnational level, green finance projects are significantly more effective than traditional development finance in reducing CO2 emissions, especially in regions with mature financial systems and strong policy support. Traditional finance may have limited or even negative environmental effects unless explicitly directed toward green objectives.

Taken together, the existing evidence shows that green-oriented finance can curb CO2 emissions, especially where governance and market depth support absorption, yet most subnational findings are China-centric, many designs are correlational, and "climate" vs. "traditional" finance is often measured with coarse, donor-provided labels. Furthermore, most of previously cited sttudies are focused on green finance, not climate finance that is especially designed toward adaptation or mitigation purposes, whereas green finance is more widely "environment related". This leaves four central gaps: (i) limited external validity at fine geographic scales, (ii) weak causal identification of dynamic, project-level effects and spillovers,

(iii) insufficient separation of mechanisms (mitigation vs. adaptation) and measurement error in climate tagging, and (iv) limited insights looking especially at climate finance.

This study addresses these gaps by assembling a global ADM2 panel that links project-level development and climate finance (Bomprezzi et al. 2024) to high-resolution emissions outcomes, and by using ClimateFinanceBERT (Toetzke, Stünzi, and Egli 2022) to validate climate content beyond donor labels. Leveraging modern staggered DiD estimators, I identify the dynamic effects of climate and traditional finance on CO2 levels and intensity. The project granularity enables clean contrasts between mitigation and adaptation portfolios, while text-audited tagging provides the first large-scale test of greenwashing/mislabeling in aid data. By delivering the first worldwide, causally identified, subnational assessment of how climate vs. traditional development finance shape emissions, this study moves the literature from suggestive correlations toward place-based evidence.

I leverage spatially explicit project data and a modern identification strategy to estimate the causal effect of climate finance on local pollution. This design combines (i) staggered adoption DiD (Callaway and Sant'Anna 2021) to identify average dynamic effects and (ii) dose–response analyses that compare dynamics across post-treatment exposure bins and via continuous doses at the ADM2–year level. This two-pronged approach distinguishes targeting from treatment intensity and is well-suited to the lumpy, time-varying nature of climate finance.

# Data

I construct an ADM2–year panel by integrating two sources. First, I use the GODAD project database (Bomprezzi et al. 2024). In order to get know whether or not projects in this datset were climate related, I use ClimateFinanceBERT (Toetzke, Stünzi, and Egli 2022) to classify projects from GODAD into climate-relevant or not and for those that were climate relevant, into adaptation, mitigation or environment categories. It allows me to get a description-based estimation of overall climate finance. For financing, I prioritize disb_loc_evensplit and substitute comm_loc_evensplit when the former is unavailable. Second, I merge environmental outcomes from EDGAR, aggregated to the ADM2 level by area-weighted averaging.

For each category, the adoption year for ADM2 $j$ is defined as the first year with a strictly positive monetary amount:

$g_j \equiv \min\{t : \text{ Amount}_{jt} > 0\}$.

Post-treatment exposure is the cumulative amount from the adoption year to the end of the panel:

$\text{Exposure}_j \equiv \sum_{t \geq g_j} \text{Amount}_{jt}$.

I partition treated ADM2s into intensity bins given by the empirical quantiles of $\text{Exposure}_j$. For descriptive cross-country figures, country–year totals are computed directly from GODAD by aggregating project amounts to the country–year level.

For the outcome variable, I build a global ADM2–year panel of territorial CO  emissions by aggregating the EDGAR 0.1° "CO  TOTALS" rasters (2000–2022) to second-level administrative boundaries. ADM2 polygons are loaded from GADM, validated and harmonized (ISO3, ADM1/ADM2 names, stable GID), and re-projected to WGS84. For each year, I read the corresponding EDGAR raster (netCDF/GeoTIFF), ensure CRS consistency, and for efficiency, crop by country envelope before extracting area-weighted sums over each ADM2. The resulting tonnes are stored and cached. Note that these ADM2 values are *model-based allocations* of national/sector totals onto grid cells rather than native administrative measurements; precision is therefore higher where emissions come from mapped point sources (power/industry) and lower for diffuse sectors, and results can be sensitive to boundary definitions and small polygons, issues that are mitigated with exact area weighting and explicit versioning of both EDGAR and GADM.

# Empirical Strategy

Let $Y_{jt}$ denote log pollution in ADM2 $j$ and year $t$. We estimate group-time average treatment effects using the **CS** estimator with not-yet-treated controls:

$$ATT(g, e) = \mathbb{E}[Y_{jt}(1) - Y_{jt}(0) \mid g_j = g, , t - g = e],$$

and aggregate to dynamic event-time profiles with simultaneous confidence bands. To study heterogeneity by dose, we compute post-treatment totals (commitments/disbursements) and re-estimate dynamics within dose bins. This recovers a policy-relevant "more money means larger effects?" gradient while preserving the staggered-adoption identification.

## Sample description

Table 1: Panel overview (ADM2–year).

| Statistic | Value |
|---|---|
| Obs. (ADM2–year) | 878 669 |
| ADM2 units | 38 203 |
| Years | 23 |
| Year span | 2000–2022 |
| Ever treated ADM2 | 4 736 |
| Never treated ADM2 | 33 467 |
| Share treated by last year | 12.4% |

One can observe from Table 1 that the panel is large and nearly balanced, 38,203 ADM2 units observed over 23 years (2000–2022) yielding 878,669 unit–years, so missingness is minimal and pre-trend and medium-run dynamics are feasible. Only 12.4% of ADM2s are ever treated by 2022 (4,736 units), which provides abundant never- or not-yet-treated controls well suited to modern staggered DiD designs, while warranting overlap diagnostics and, if necessary, reweighting or trimming to common support. Staggered adoption and the definition of treatment as the first positive climate amount naturally motivate dynamic event-study ATTs alongside dose–response analyses based on cumulative post-treatment exposure. For clarity, note that the reported "share treated by last year" is the fraction ever treated by the final year, not the fraction treated in that year. Given the sample size and number of treated clusters, average effects should be precisely estimated.

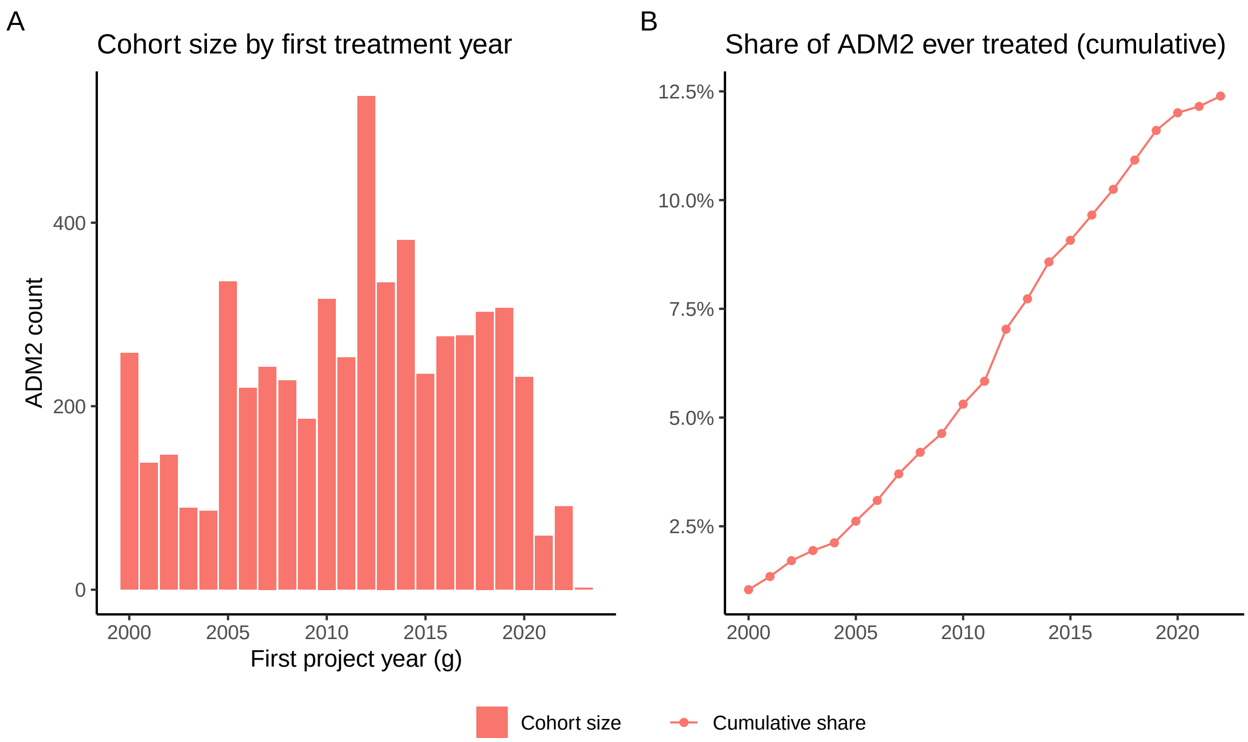
Table 2: Descriptive statistics for log $CO_2$ outcomes: overall and by treatment status.

| Group | N | Mean | Sd | P10.10% | P50.50% | P90.90% |
|---|---|---|---|---|---|---|
| **Overall** | **878669.000** | **10.558** | **2.072** | **8.308** | **10.412** | **13.195** |
| Never treated | 769741.000 | 10.441 | 2.008 | 8.264 | 10.312 | 12.983 |
| Ever treated | 108928.000 | 11.386 | 2.312 | 8.744 | 11.148 | 14.511 |

In terms of outcomes, Table 2 shows that the distribution of log CO  is clearly right-shifted for ever-treated ADM2s relative to never-treated ones: the mean gap is 0.945 log points, implying treated ADM2s exhibit about 2.57× higher CO  levels on average $e^{0.945} \approx 2.57$, with a median gap of 0.836 log points ≈ 2.31×. Dispersion is larger among treated units (SD 2.31 vs 2.01), and the displayed percentiles are uniformly higher ( $P10 : +0.48 \quad log \approx +62\% \quad P90 : +0.212 log \approx +24\%$), suggesting a broad rightward shift rather than just upper-tail differences. Cross-sectionally this is consistent with selection: treated ADM2s are likely more urban/industrial or otherwise systematically different. A pooled standardized mean difference of ≈ 0.46 (using group SDs and sample sizes) indicates moderate imbalance that should be addressed. Substantively, these gaps underscore the need to rely on within-ADM2 identification (unit and year fixed effects), probe parallel trends with event-study leads.

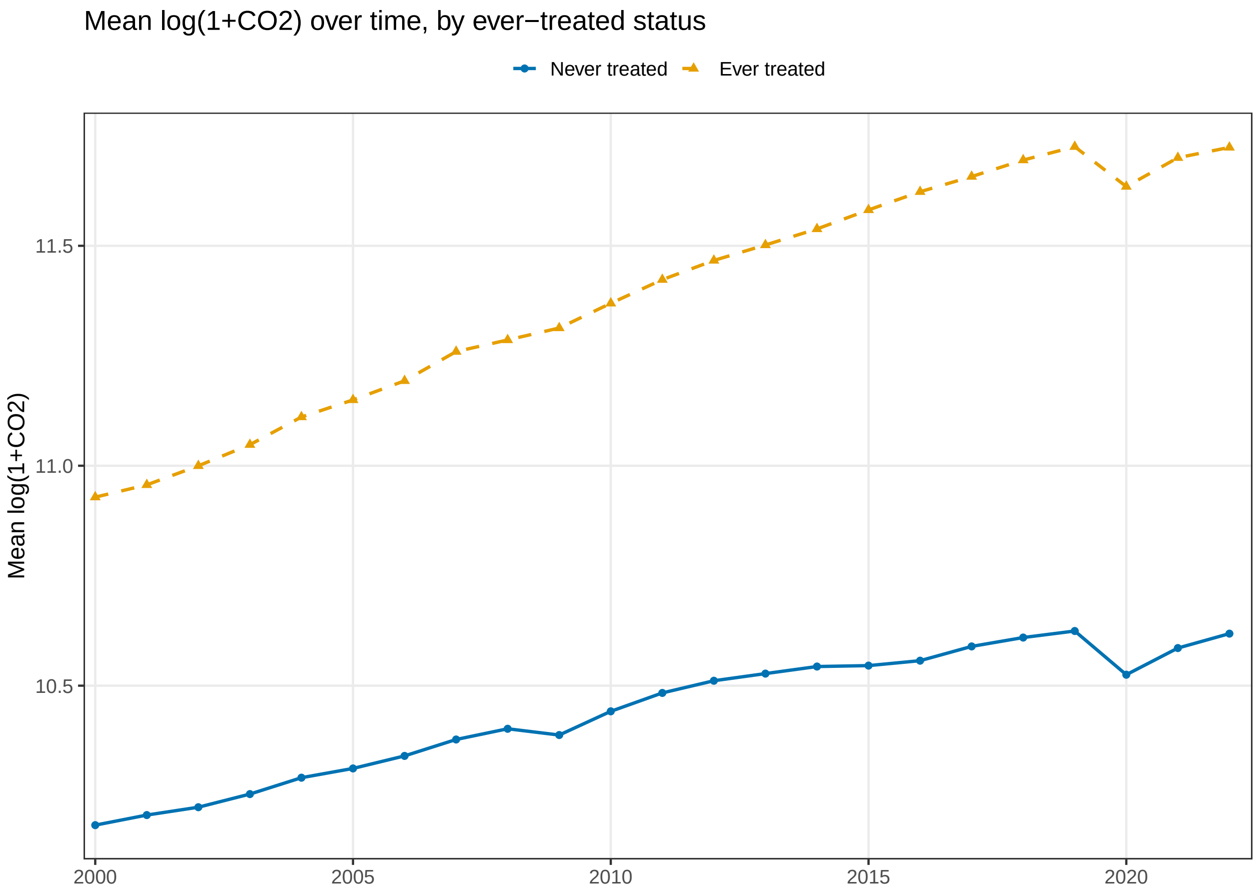
According to Figure 1, the adoption profile is clearly staggered and concentrated in the early-to-mid 2010s: Panel A shows modest early cohorts, a marked ramp-up around 2011–2016, and thinning cohorts thereafter (partly right-censoring near the panel end), while Panel B's S-shaped curve plateaus at ≈ 12–13% of ADM2s by 2022. Combined with earlier descriptives, treated ADM2s are systematically more CO -intensive, this pattern is consistent with selective rollout toward higher-emitting, more urban/industrial areas rather than random diffusion. Empirically, these features favor staggered event-study designs with rich fixed effects, but they also call for explicit pre-trend tests (anticipation and dose–response specifications using post-treatment exposure.

In Figure 1, the unadjusted series show a stable level gap, ADM2s that ever receive climate projects have consistently higher $log(1 + CO_2)$, but the two groups move along very similar secular paths. From 2000 to the early-mid 2010s both lines trend upward with comparable slopes, and there is no obvious pre-period divergence, which is reassuring for the parallel-trends requirement. Because "ever treated" is defined over the whole sample, these differences are descriptive, not causal, and reflect composition (future-treated units are included in pre-years). Accordingly, my identification will rely on within-ADM2 variation and dynamic event-study leads to test for anticipation. I will also probe heterogeneity and spillovers, but




Note: g denotes the first year an ADM2 receives a positive climate amount (adaptation or mitigation). Panel A shows the count of ADM2s by g (years >= 2000). Panel B plots the cumulative share of all ADM2s with g <= year; the denominator includes never−treated ADM2s. Percent axis in Panel B refers to this share. Missing g implies never−treated; cohorts are right−censored at the panel end.

Figure 1: Adoption dynamics at ADM2 level: (A) cohort size by first treatment year; (B) cumulative share of ADM2 ever treated.




Note: Series are unweighted ADM2 means of log(1+CO2) from EDGAR aggregated to ADM2.
'Ever treated' is defined over the full sample (ADM2s that ever receive a positive climate amount by 2022),
so pre−treatment years for that group include future−treated ADM2s. These differences are descriptive and not causal;
sample sizes vary by year. Legend encodes both color and linetype for print accessibility.

Figure 2: Unadjusted means of $\log(1+CO_2)$ by treatment status over time.

the aggregate pattern here is broadly consistent with a DiD design: different levels, similar trends.

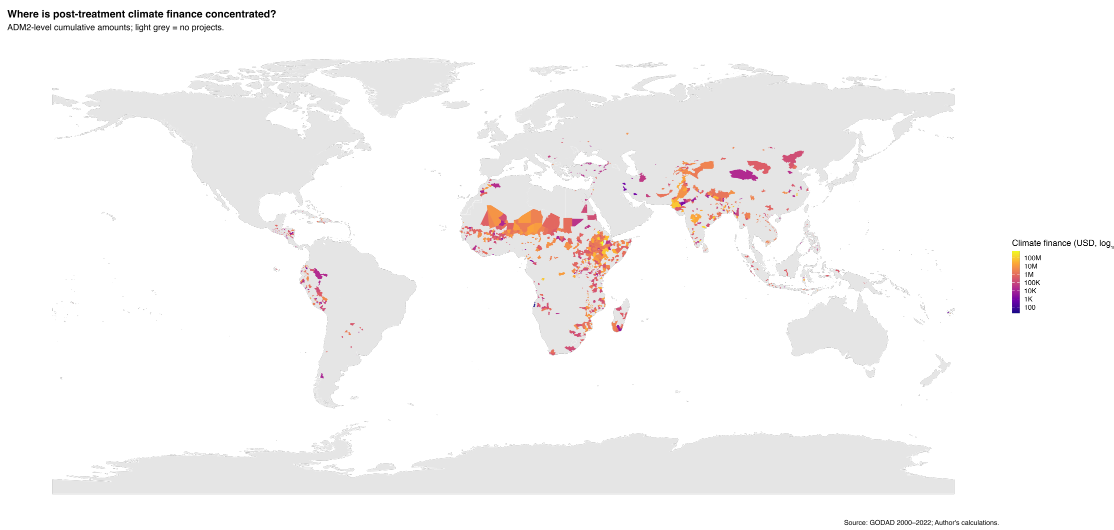



Figure 3: ADM2-level cumulative climate finance (USD, log scale). Light grey = no projects.

Figure 3 shows a highly clustered footprint of climate-project exposure at the ADM2 level with a dense concentrations across Sub-Saharan Africa (Sahel, Great Lakes, coastal West Africa), South Asia (Indo-Gangetic plain and Himalayan foothills), and parts of Southeast Asia, with smaller pockets in the Andes and Central America; coverage is sparse in North Africa/Middle East and essentially absent in high-income countries (by design). This spatial pattern is consistent with targeting toward populous/urbanizing and vulnerable corridors.

# Results

I identify dynamic treatment effects by comparing newly treated ADM2s to ADM2s that have not yet received a climate project in that event-time, which preserves credible counterfactuals under staggered adoption. With only 12.4% of ADM2s ever treated by 2022, the panel supplies abundant not-yet-treated controls for each cohort–event cell. Event-study leads are near zero, supporting parallel trends in the pre-period. I identify the first year in which an

ADM2 receives at least one climate project. If the GODAD file already has an ADM2 code, I use it; if not, a spatial join using the point geometry is performed.

Using not yet treated estimation allows to reduce the selection into treatment, also, as it is expected to see a decrease in CO2 emission with implementation of climate-related development projects, non-treated, bordering ADM2 areas shall also be benefiting from treatment. This allows a downward bias to the results that will be presented after.

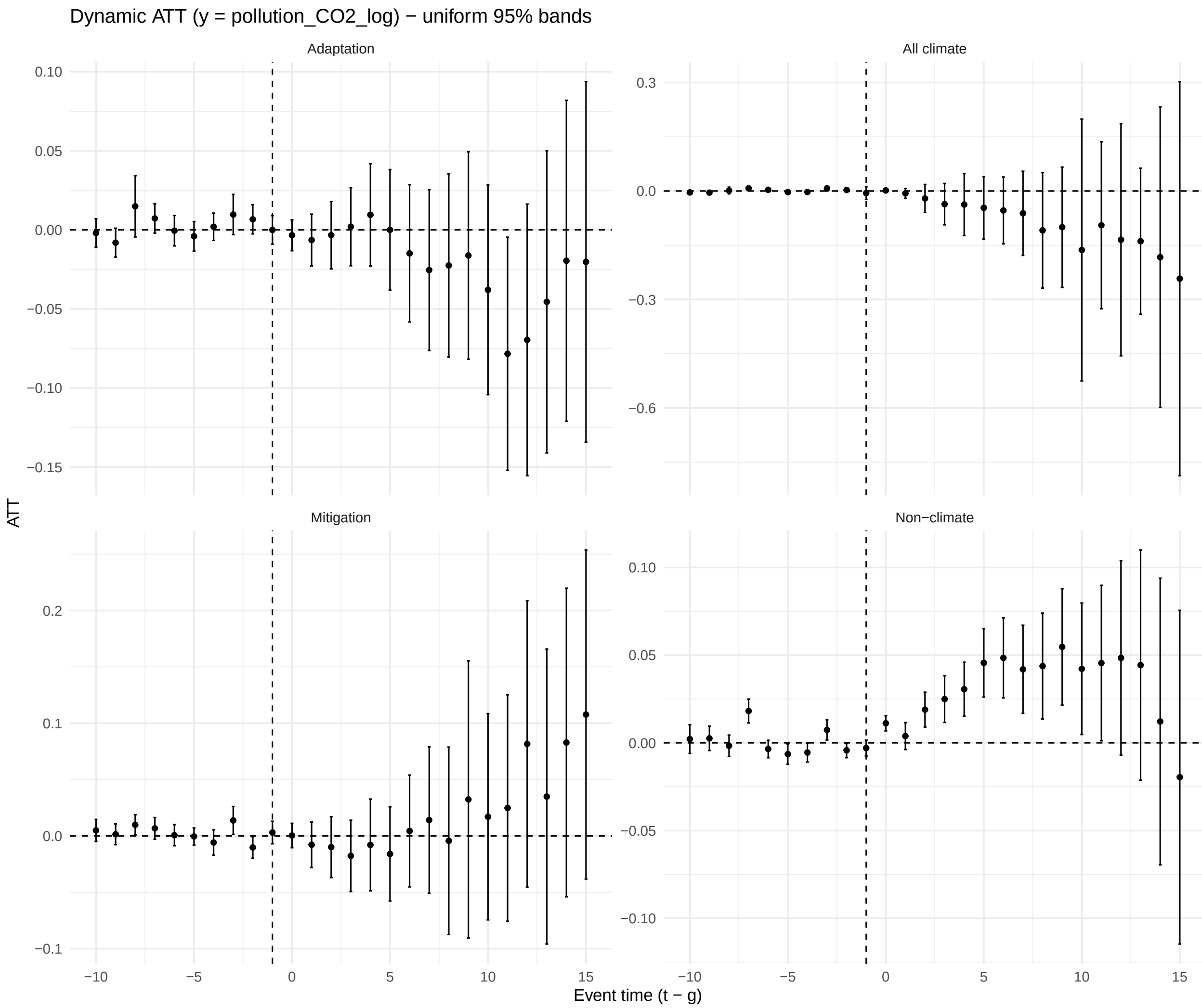



Figure 4: Dynamic treatment effects of aid adoption on local CO2 emissions (log). Event-time estimates (t−g) from Callaway–Sant'Anna DiD using not-yet-treated controls, doubly-robust estimator, and uniform 95% simultaneous bands (multiplier bootstrap; clustered at ADM2).

The Figure 4 dynamic event-study estimates (uniform 95% bands) indicate no evidence of anticipatory behavior: pre-treatment coefficients are small and statistically indistinguishable from zero across all specifications, supporting the parallel-trends assumption. Aggregating categories, all climate projects yield a modest negative long-run effect, whereas non-climate aid shows small positive effects, in line with the scale effects of general development spending. Furthermore, aid boosting economic activity and therefore CO2 emission is an expected results, its magnitude might be reduced due to spillover effect. Indeed, bordering region might also benefit from treatment in terms of economic activity but also in terms of emission spreading. This could increase control group emission as well, reducing the effect of development projects.

# Heterogeneity

## Event study by post-treatment dose bins

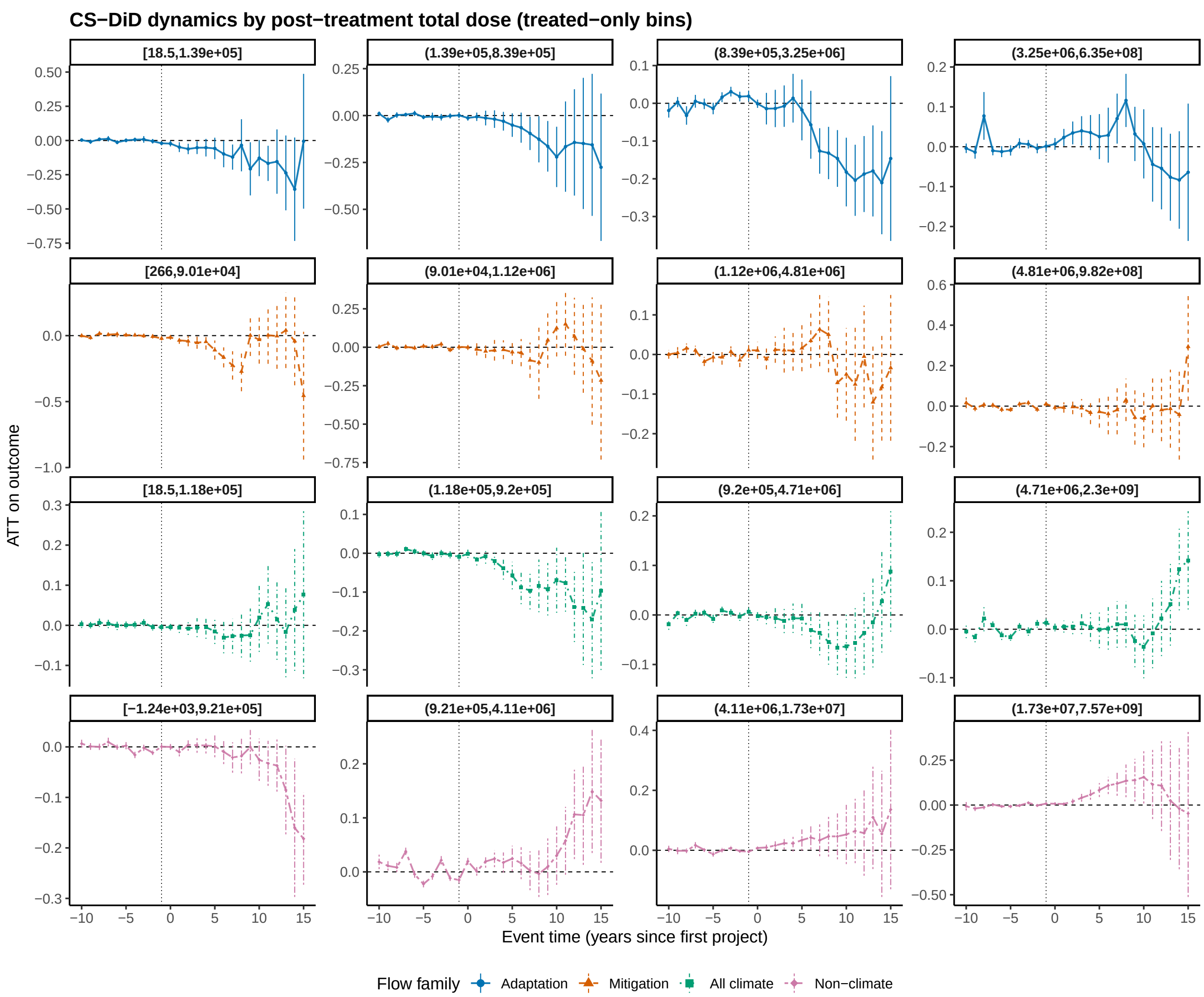



Figure 5: Event-study by post-treatment total dose — stratified bins (treated-only cuts).

**Interpretation.** Higher post-treatment exposure is associated with larger post-event estimates, consistent with a **dose–response** relationship. Pre-treatment coefficients remain centered at zero across bins. As we can see in Figure 5, event-time dynamics are shown separately by post-treatment exposure bins (treated-only cuts). Across families, pre-treatment coefficients hover near zero with wide, overlapping bands, supporting the parallel-trends assumption. Post-treatment, the **climate portfolios** display the expected *intensity gradient*: in mid- and high-dose bins, adaptation and (to a lesser extent) mitigation tilt negative in the medium run (roughly e [5,12]e [5,12]e [5,12]), with several bins showing $-0.1$ to $-0.3$ log-points, while low-dose bins are small and often indistinguishable from zero. By contrast, **non-climate finance** tends to move **positive** at higher doses and longer horizons (late eee), consistent with scale effects from general activity. The **all-climate** aggregate mirrors these patterns—modest declines that strengthen with dose—although extreme bins occasionally exhibit noise and reversion, reflecting fewer treated units and wider uniform bands. Overall, the figure indicates (i) no systematic anticipatory trends; (ii) climate projects reduce local CO  more where cumulative funding is larger; and (iii) conventional development finance can partly offset these gains at high intensity. We caution that several bin-specific profiles have broad confidence intervals, so inference should emphasize the cross-bin monotone pattern rather than any single coefficient, and we verify robustness to alternative binning rules and the exclusion of influential countries in subsequent checks.

## Event-study per climate class

```
            class     g     N
           <char> <int> <int>
  1:          500  2001  1202
  2:          500  2002   773
  3:          500  2003   760
  4:          500  2004  1251
  5:          500  2005   984
 ---
224: Wind-energy  2013     5
225: Wind-energy  2016     5
226: Wind-energy  2017    14
227: Wind-energy  2019     5
228: Wind-energy  2020     7
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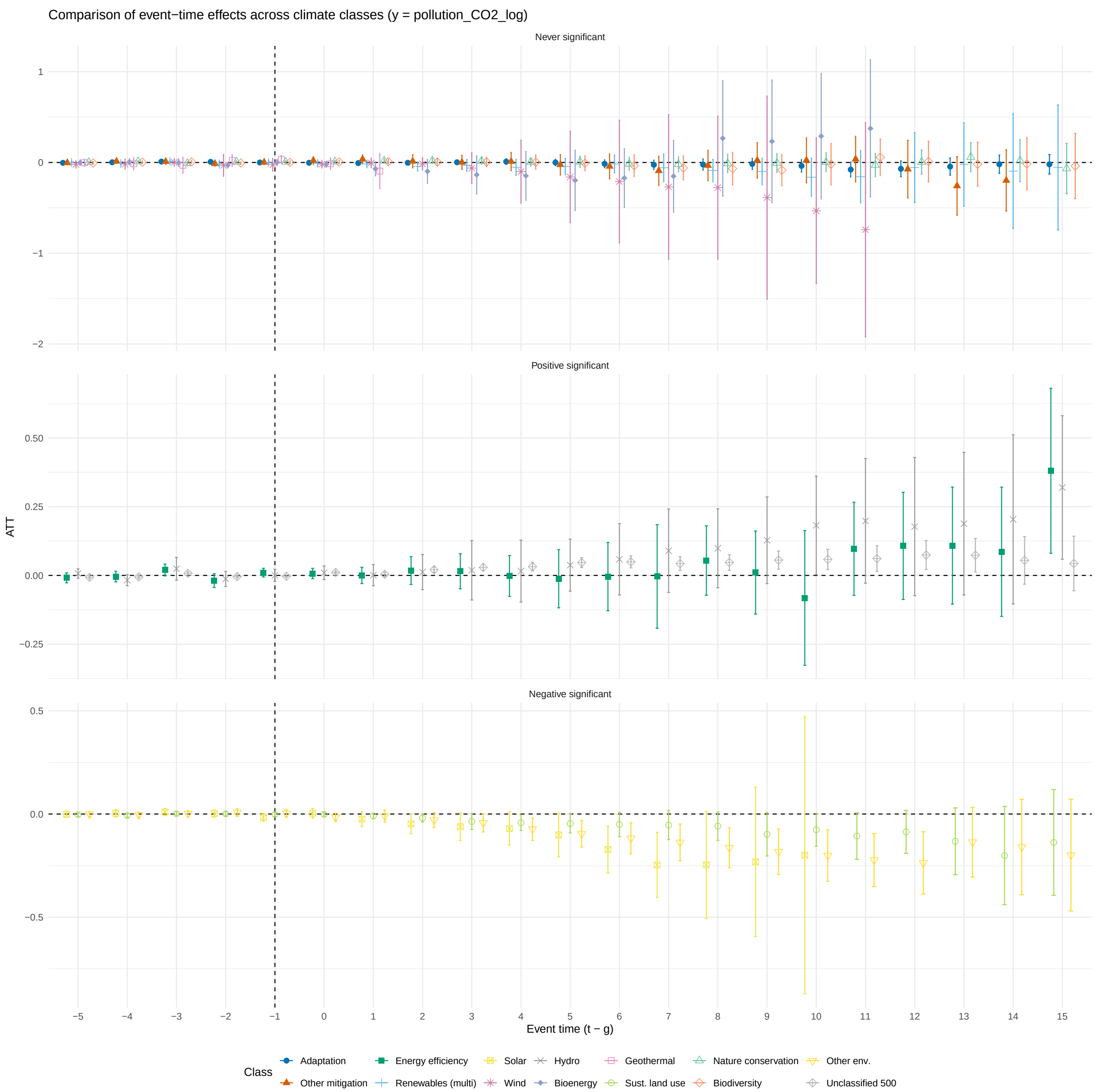


Figure 6: Comparison of event-time effects across climate classes

# Robustness Checks

## Placebo test

To probe identification, I implement a placebo event–study that advances the cohort year by four years $g^{pl} = g - 4$ and re-estimates dynamic effects on the true pre-treatment sample. Results are displayed in Figure 7 . Across all four categories (Adaptation, Mitigation, All climate, Non-climate), placebo ATTs are close to zero around the pseudo-event time and the uniform 95% bands typically cover zero throughout the window, with no systematic departures. The mild drifts with widening intervals in far "post" leads are consistent with thinning support at long leads rather than genuine effects. Because any non-zero placebo effect would reflect differential pre-trends, these results provide no evidence of anticipatory dynamics or pre-trending, lending credibility to the parallel-trends condition underpinning our baseline estimates.

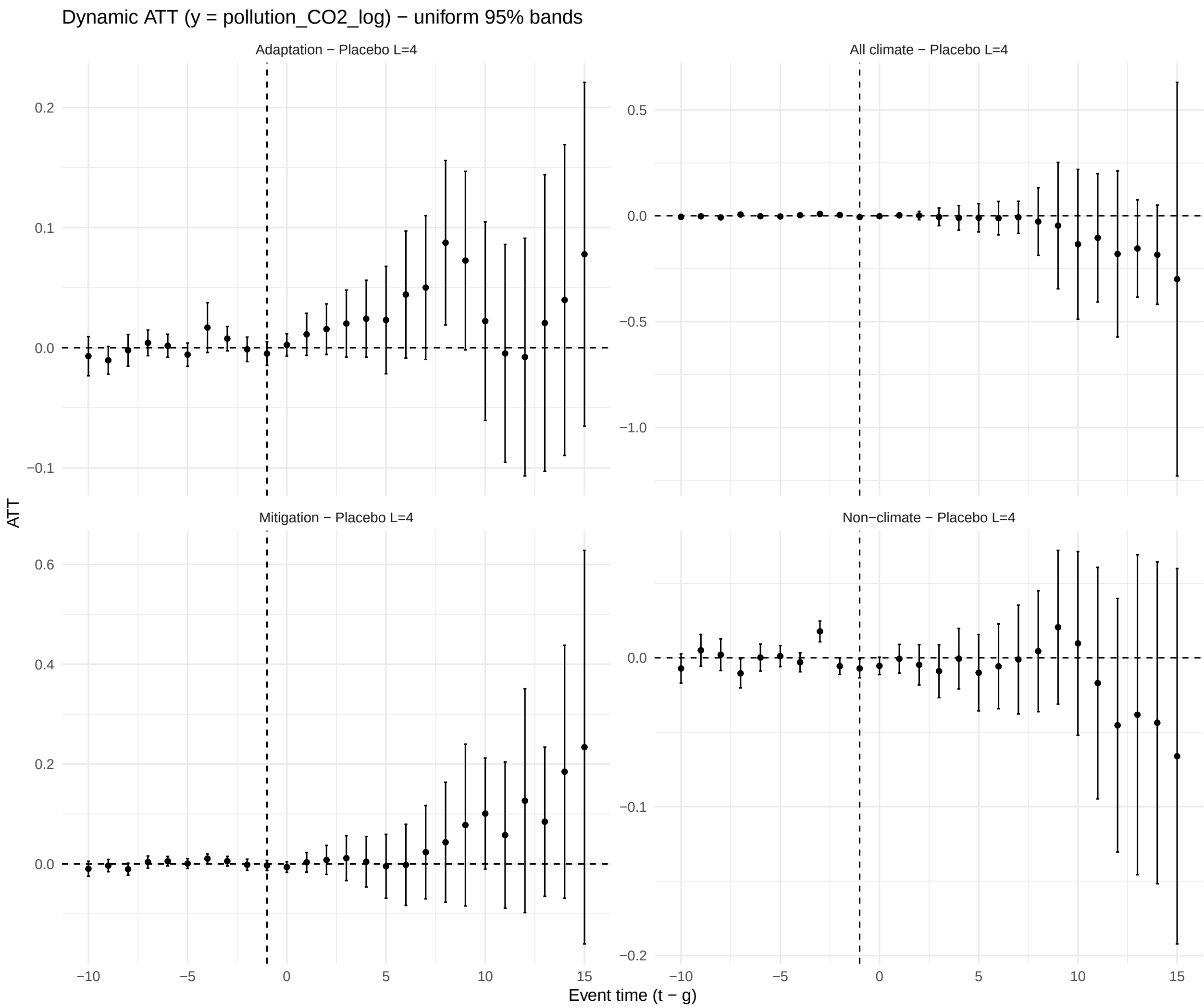


Figure 7: Dynamic treatment effects of aid adoption on local CO2 emissions (log). Event-time estimates (t−g) from Callaway–Sant'Anna DiD using not-yet-treated controls, doubly-robust estimator, and uniform 95% simultaneous bands (Placebo).

## Excluding China

As one can see in Table A1, chinese adm2 region are overrepresented in top emission treated adm2. To assess the sensitivity of the results, I will exclude China from the regression to stress the results.

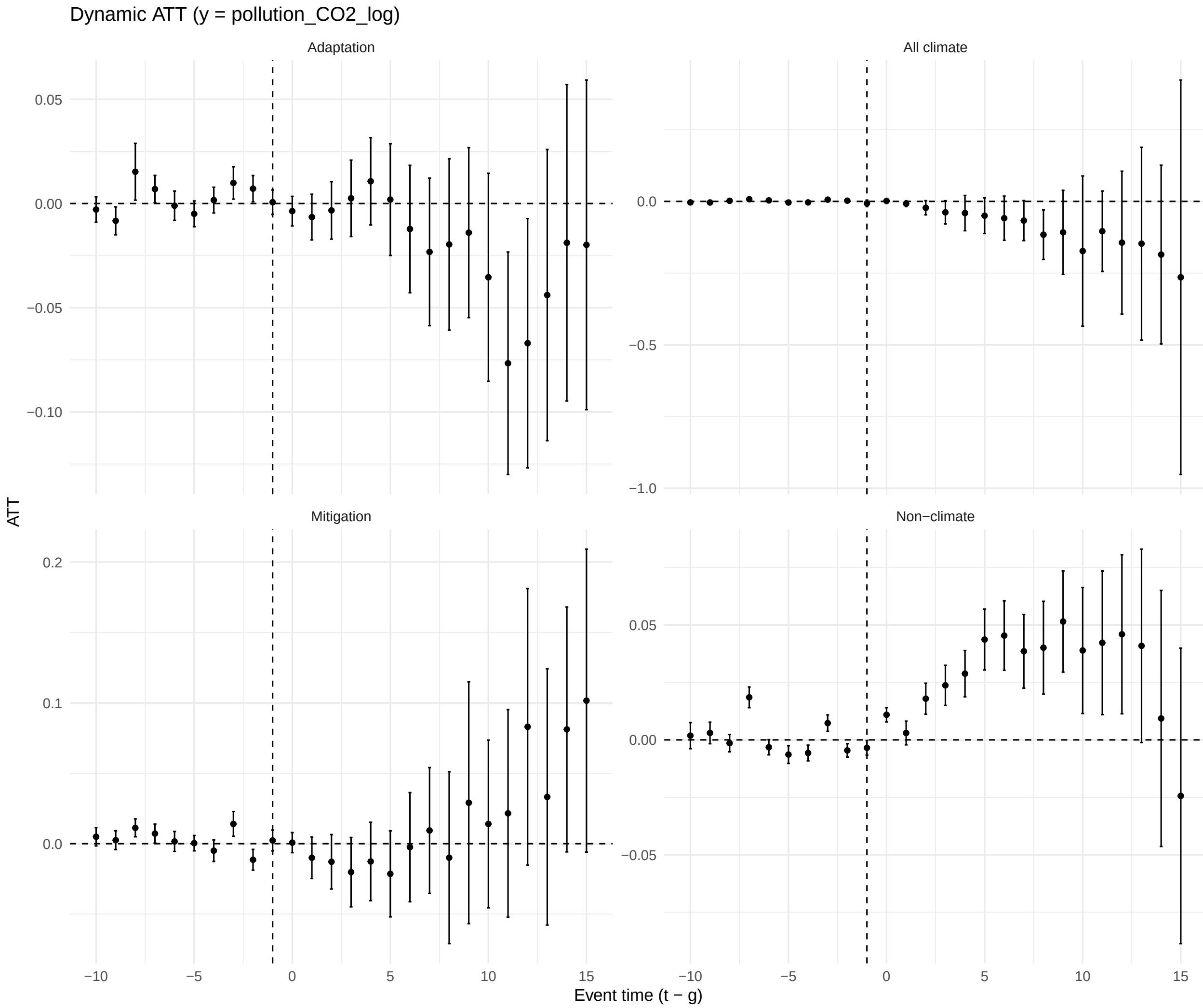


Figure 8: Dynamic treatment effects of aid adoption on local CO2 emissions (log, China excluded)

As we can see in Figure 8, removing chinese observations does not change the results. It even improves it if we look at adaptation aid, where one can see a significant drop in emission.

## Intensity

As well for the intensity, the methodology is reproduced to stress the results in excluding chinese adm2 in Figure 9.

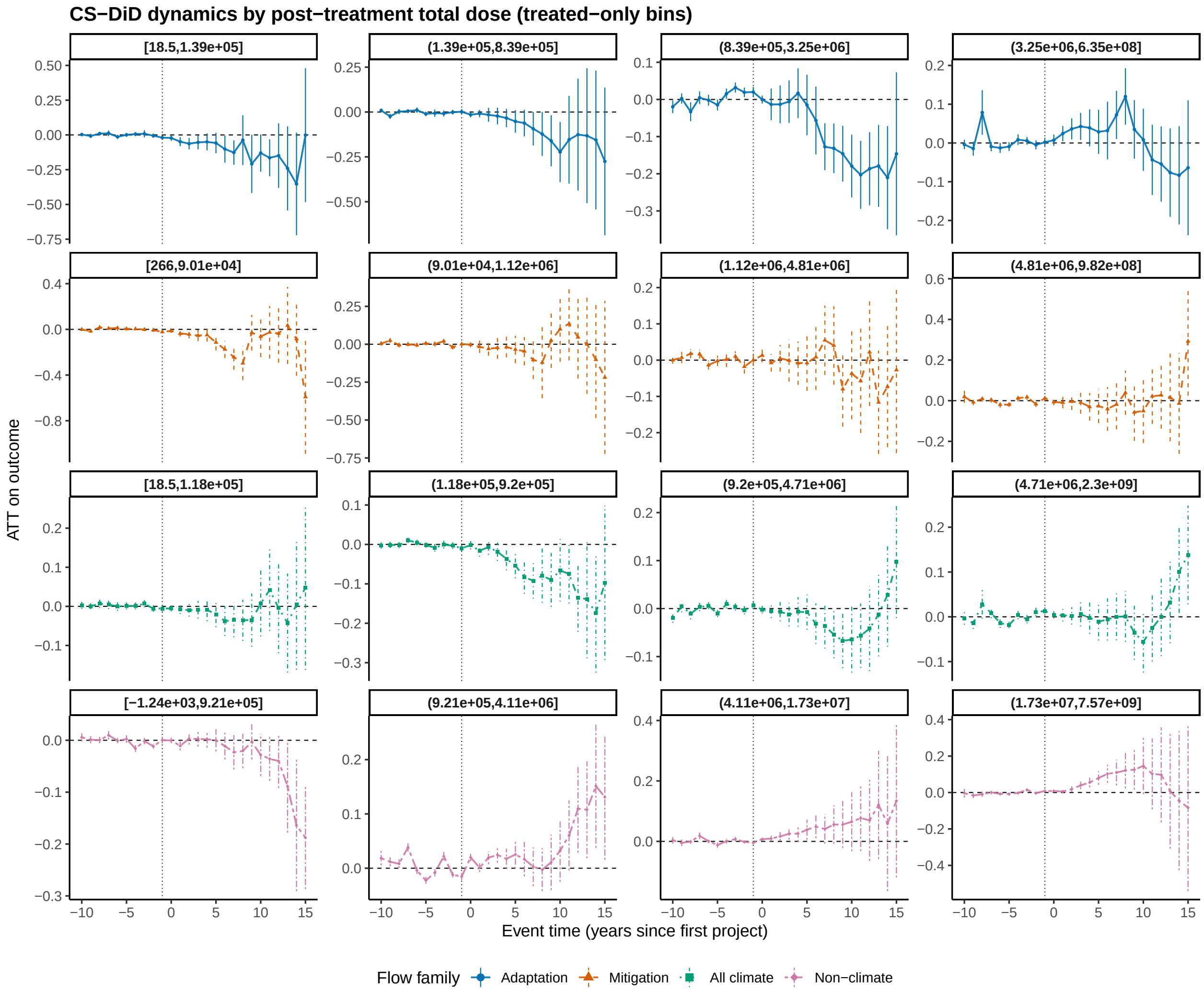


Figure 9: Event-study by post-treatment total dose — stratified bins (treated-only cuts, China excluded).

Excluding China leaves the core patterns intact and, if anything, clarifies the dose–response. Pre-treatment coefficients remain close to zero across bins, supporting parallel trends. On the treatment side, climate portfolios continue to show medium-run declines in local log-CO that strengthen with cumulative exposure: attenuation at very low doses, progressively more negative ATTs in the mid/high-dose bins, and a similar profile for the all-climate aggregate. Non-climate finance again tilts positive at longer horizons in higher-dose bins, consistent with activity scale effects. Magnitudes are modestly smaller and several bands are wider, expected given the drop in sample size and the removal of a high-leverage country, but the cross-bin monotone pattern (more climate money implies larger emissions reductions) survives. Overall, this figure demonstrates that the main results are not driven by China: there are no systematic anticipatory trends, climate funding reduces emissions where it accumulates, and conventional aid can offset these gains at high intensity. We therefore interpret the China-excluded estimates as a robustness check that reinforces the global findings rather than a substantively different story.

# Mechanisms

To support the idea of non-climate related projects fostering development and therefore emission, I collected nighttime lights (NTL) yearly averaged at adm2 level in treated areas. I proxy local economic development with NTL for two reasons. First, a large literature shows that variation in NTL captures intensive-margin economic activity and infrastructure at fine spatial scales, especially where income data are sparse (J. V. Henderson, Storeygard, and Weil 2012; Chen and Nordhaus 2011; Pinkovskiy and Sala-i-Martin 2016). Second, NTL reacts contemporaneously to investments in roads, electrification, and urban services, giving a timely marker of development intensity at ADM2. The ex-ante hypothesis was therefore: non-climate aid (and some climate-related investments) should increase local development, captured by NTL, and, at least in the short run, also increase CO  emissions, via higher energy use, transport, construction, and operation of industrial/process activities.

Figure 10 (event studies on $log(1 + VIIRSNTL)$) shows (i) broadly flat pre-trends and (ii) at most modest post-t dimming of NTL for climate and non-climate projects in the *aligned* sample, particularly in already bright places (see Figure 11) . Taken at face value, this pattern appears inconsistent with the "more project equals more light" channel. However, it is consistent with two theory-driven mechanisms:

1. LED/spectral-bias and dimming: Many projects retrofit public lighting and street infrastructure with LEDs or install smart dimming controls for energy efficiency. VIIRS-DNB is relatively less sensitive to blue-rich white LEDs and to reduced operating hours, so measured radiance can decline even when lumen output or activity does not (X. Li et al. 2017; Kyba et al. 2017; Levin et al. 2020). In parallel, improved connectivity and reliability can raise daytime activity, fuel use, and process/transport emissions, explaining a raise in CO2 with NTL near zero.

2. Daytime/process intensity with limited lighting: Industrial parks, cement and extractive activities, logistics hubs, and road programs primarily lift daytime or process loads (EDGAR industry/transport sectors) while adding little additional night lighting. This generates a high CO2 with little NTL response. Furthermore, in areas that are already highly illuminated, the likelihood of receiving large infrastructure investments (e.g., road construction) is greater. However, such projects can reduce observed night-time radiance when residential or commercial land uses, typically brighter at night, are replaced by transport infrastructure that emits less sustained light.

```
--- SAMPLE SIZE (obs = ADM2×year) ---

Original outcome (pollution_CO2_log): 878 669

VIIRS night-lights   : 566 592

Common keys: 382 020 | Only CO2: 496 649 | Only VIIRS: 184 572

Aligned dataset: 382 020 obs, 38202 ADM2
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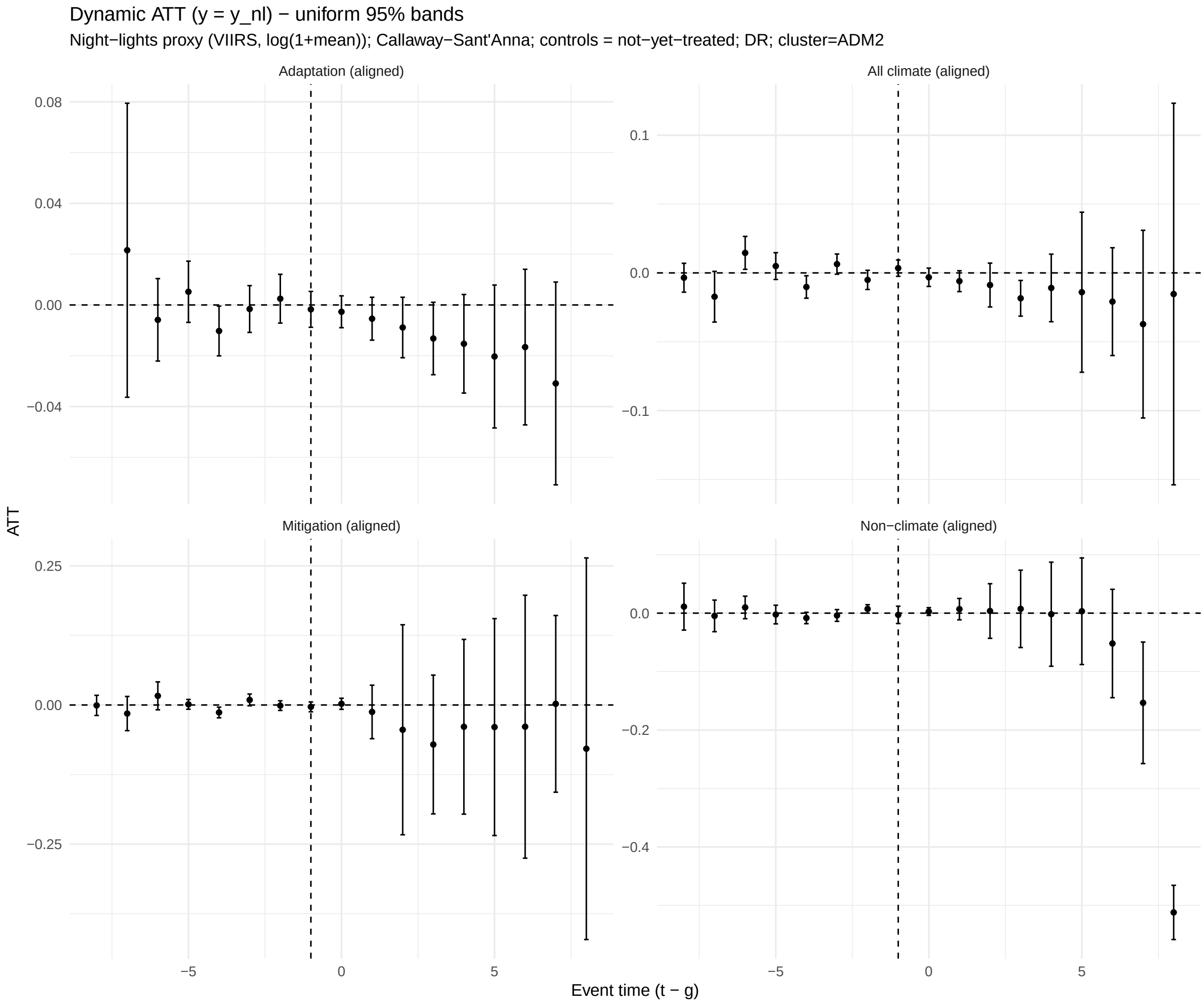


Figure 10: Dynamic treatment effects of aid adoption on NighTime Lights (log, ADM2)

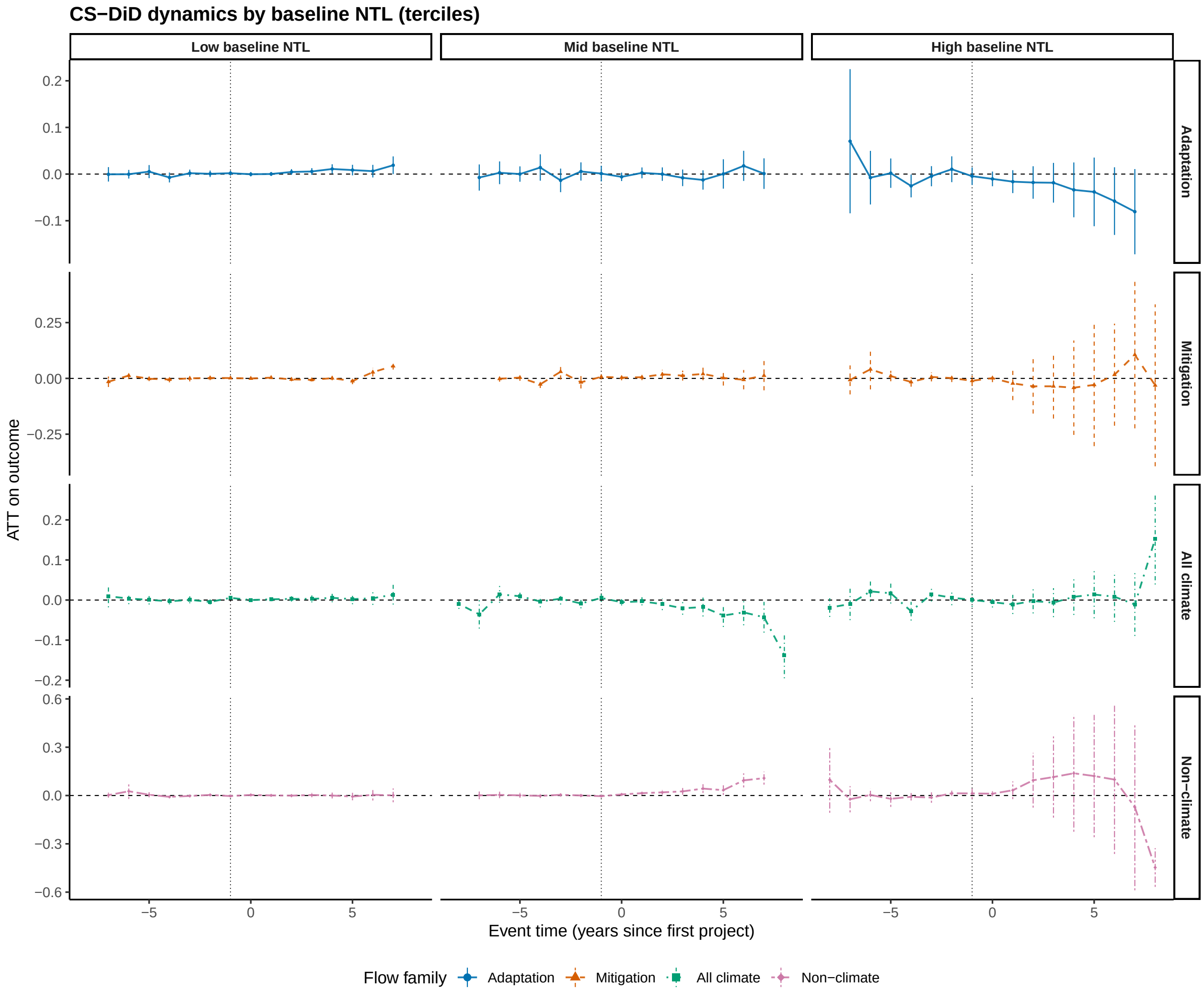


Figure 11: CS-DiD dynamics in night-time lights by baseline brightness (terciles)

Figure 11 directly speaks to mechanism (1): when we split the sample by baseline brightness terciles (2013), the post-t dimming is concentrated in the high-NTL tercile, precisely where public lighting stocks are largest and LED/dimming retrofits bite most. Pre-t coefficients remain near zero across terciles, reinforcing identification. Furthermore, in mid tercile of NTL at beginning of the observed period, one can see that non climate project are increasing NTL, whereas in high NTL, the lowering of NTL is coming at the end of the observable period and might be driven by high industrial and infrastructural projects. Figure 12 illustrates this.

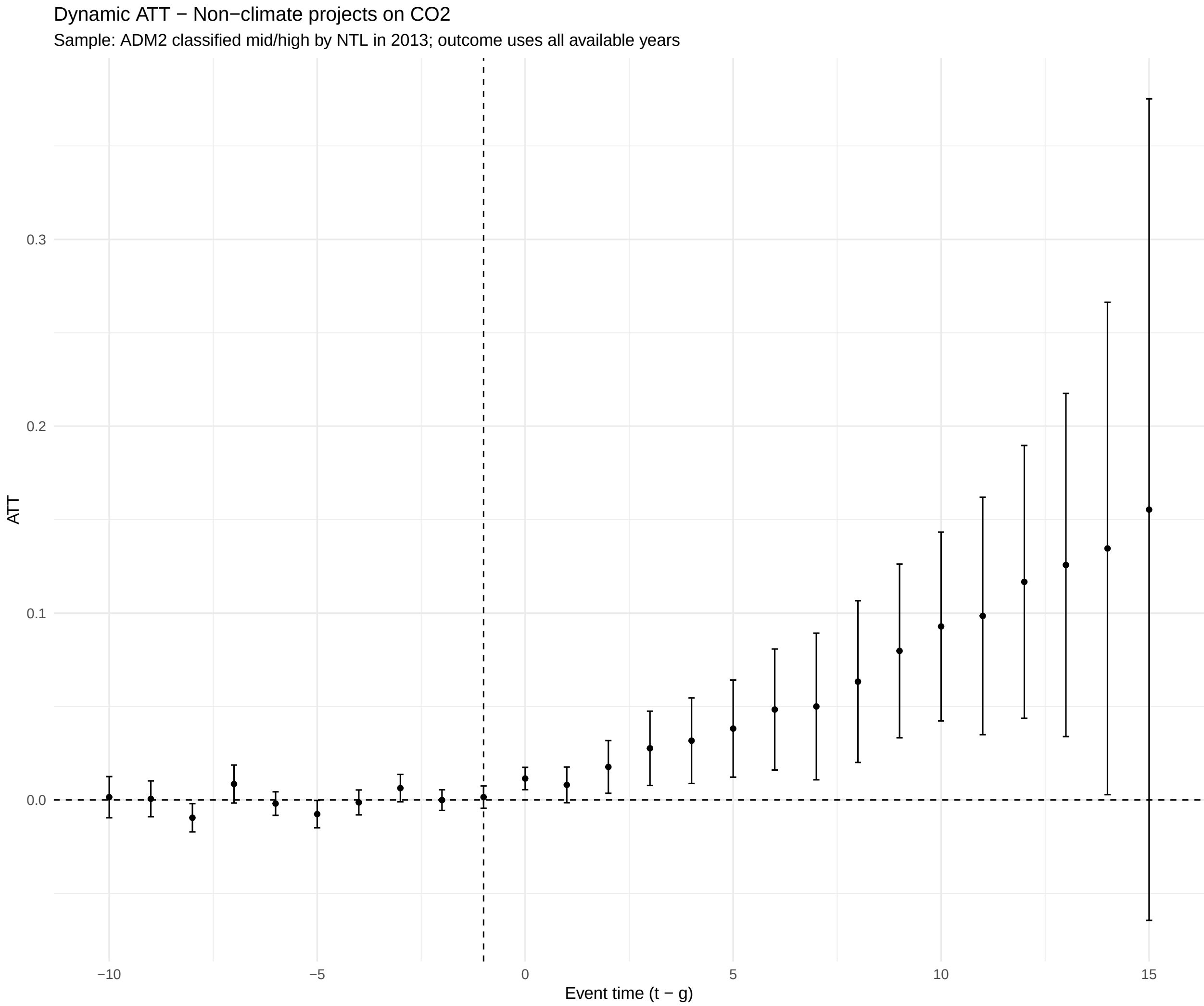


Figure 12: Dynamic ATT of non-climate projects on CO2 emissions (ADM2, mid/high NTL baseline)

Restricting the sample to ADM2 in the mid and high NTL terciles in the baseline year (2013) yields a clear post-treatment pattern for non-climate projects: CO2 emissions increase monotonically over the horizon, with flat pre-trends. Interpreting this together with our NTL results points to a development-driven mechanism in which non-climate assistance expands local economic activity, but the visibility of that expansion in night lights depends on baseline brightness and the composition of projects.

In mid-NTL areas, the project mix plausibly tilts toward development-oriented investments, housing upgrades, services, small commerce, electrification, that not only raise activity (and thus CO2) but also increase NTL, producing the expected positive association between development and measured radiance. By contrast, in already bright (high-NTL) areas, marginal brightening is harder to achieve (ceiling/saturation) and the portfolio shifts toward large transport and backbone infrastructure (roads, logistics platforms, rights-of-way). These investments scale daytime/process activity and emissions yet contribute little to sustained night radiance; they may even reduce NTL through land-use substitution (replacing brighter residential/commercial uses with less luminous transport surfaces) and through lighting efficiency changes (LED retrofits, dimming, shorter operating hours).

This mechanism reconciles the paper's core findings. First, it explains why non-climate aid raises $CO_2$ while NTL responses are muted or negative in bright places: NTL is sensitive to lighting technology and land-use composition, not just to output. Second, the heterogeneity by baseline brightness, NTL gains concentrated in mid-NTL ADM2 and attenuation in high-NTL ADM2, is consistent with selection into project types and with greater scope for efficiency retrofits where lighting stocks are large. Third, consistently flat pre-trends across specifications support the identifying assumptions of our DiD design.

Overall, the evidence indicates that non-climate assistance primarily scales up development-intensive, energy- and transport-using activity, while contemporaneous lighting and land-use changes modulate the NTL signal. This joint reading, CO 2up, NTL up in mid-NTL; $CO_2$ up, NTL flat/down in high-NTL, highlights the importance of combining NTL with sector-specific outcomes when assessing local development impacts.

## Which climate aid tends to foster development ?

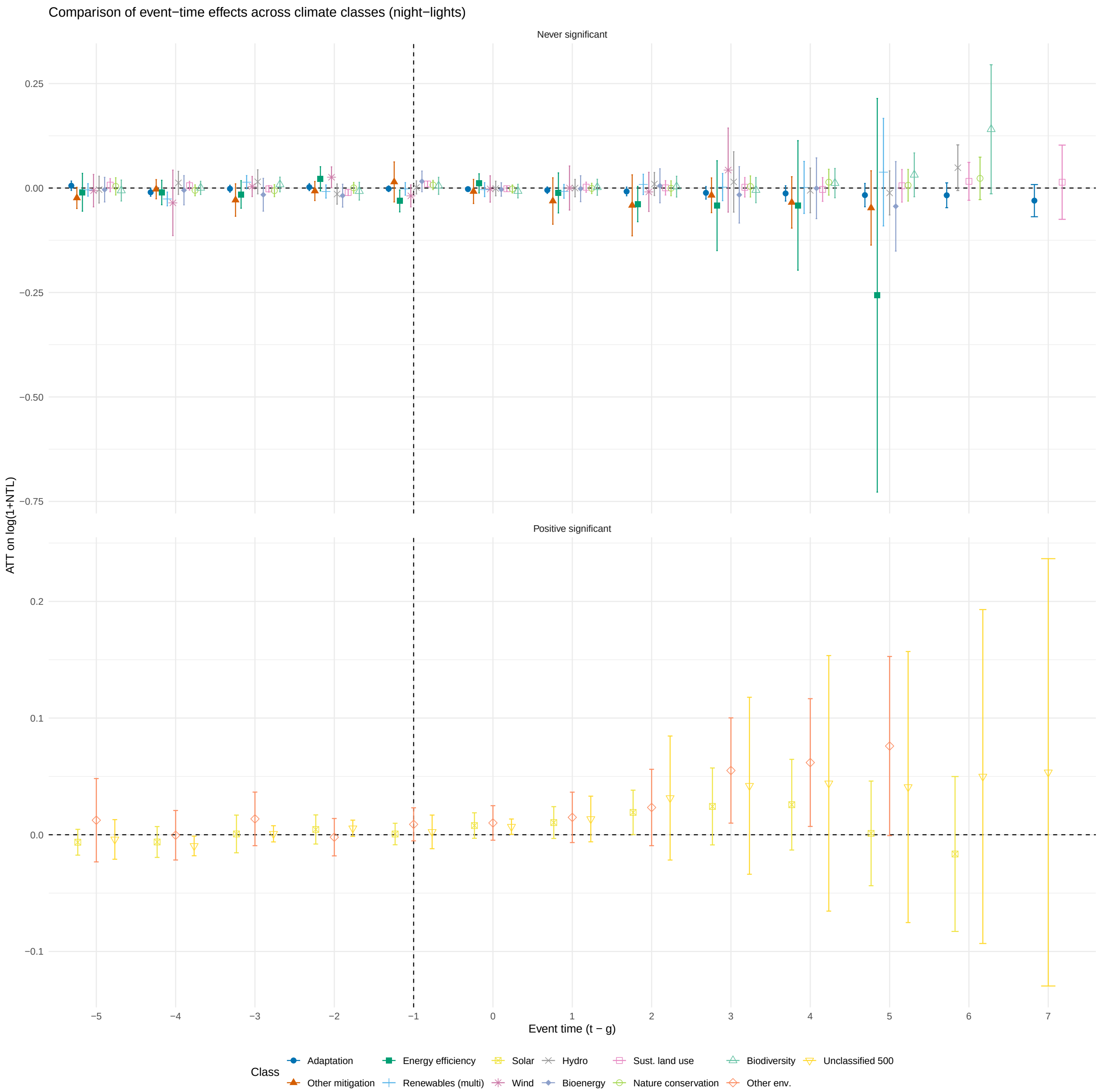


Figure 13: Event-study ATT on VIIRS night-lights (log(1+mean)) by GODAD climate classes. Facets group classes that show post-treatment uniform-significant positive, negative, or never-significant effects.

# Policy Recommendations

`Canon classes in NTL only:`

`Canon classes in CO2 only:`

```
 Geothermal-energy
Canon classes used (intersection):
 Adaptation, Other-mitigation-projects, Energy-efficiency, Renewables-multiple, Solar-en
```

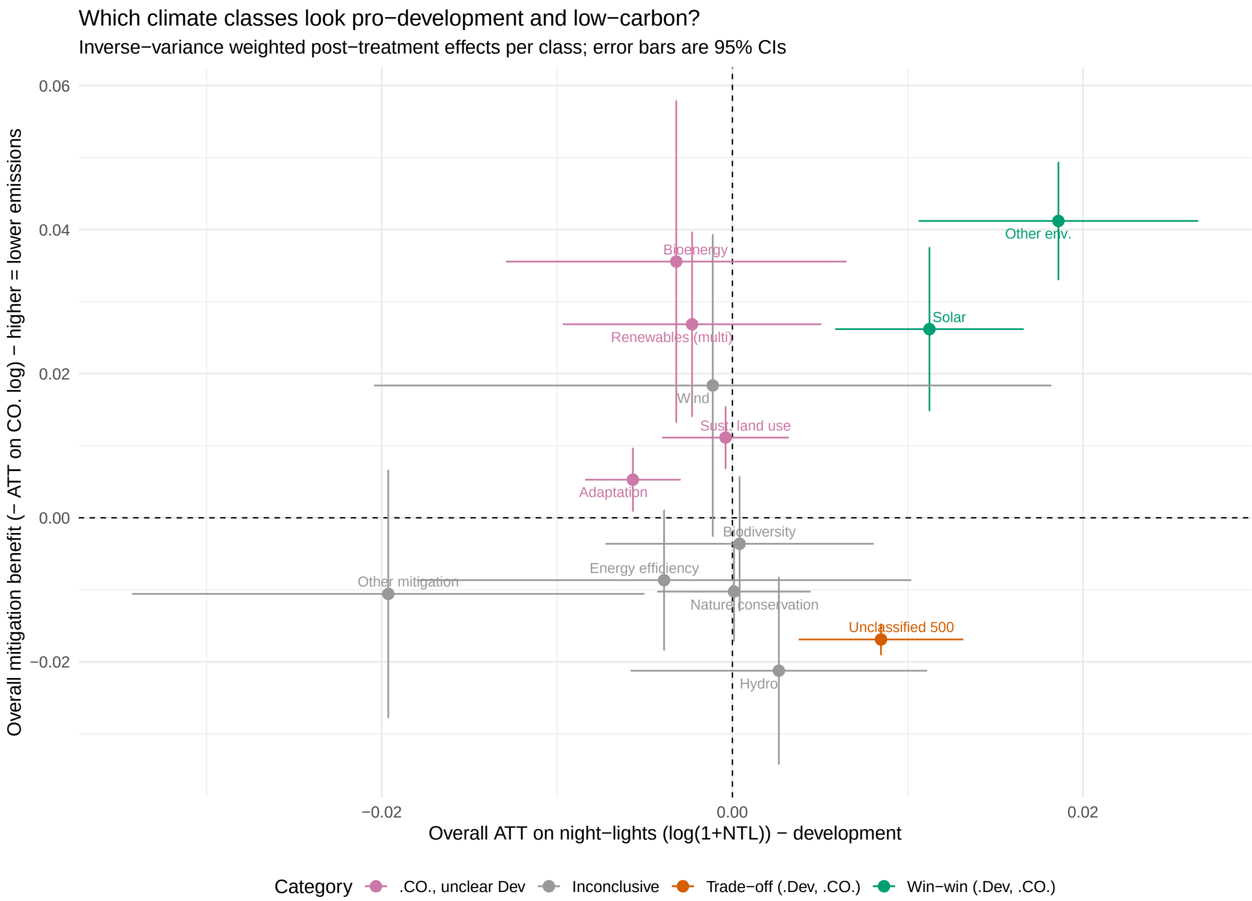


Figure 14: Development vs. GHG trade-offs by climate class. X: overall ATT on night-lights (log1p). Y: −ATT on CO  (log). Error bars are 95% CIs.

# Conclusion

This study links geocoded aid to ADM2-level CO  and shows a clear divergence: climate portfolios yield no effect or modest reductions in emissions, stronger at higher cumulative exposure, whereas non-climate aid increases emissions over the medium run. These dynamics survive placebo timing tests and the exclusion of China, with pre-treatment coefficients close to zero, supporting identification under staggered adoption.

The mechanism is development-driven. Non-climate finance expands local economic activity, as reflected in night-time lights: NTL rises in mid-brightness ADM2 where additional commerce, services, and electrification are most visible; in high-brightness areas, NTL responses become muted or negative due to LED/dimming and land-use substitution toward transport/industrial infrastructure, even as CO2 rises through daytime/process loads. Thus

"CO2 up" under non-climate aid is consistent with "development up," with NTL mapping that expansion where radiance remains informative.

Policy follows directly. Scaling climate portfolios can deliver measurable local decarbonization (particularly where absorptive capacity makes dose–response steep), while greening or retargeting non-climate spending is necessary to avoid offsetting emissions from development-intensive sectors. NTL offers a practical monitoring proxy for development-induced emissions where spectral/land-use biases are accounted for.

Limitations include sectoral heterogeneity masked by averages and potential measurement error in climate tagging. Priority extensions are sector/technology decomposition, explicit spatial spillovers.

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# Appendix

Table A1: Top 20 ADM2 by mean $CO_2$ level (EDGAR-aggregated).

| Rank | Country | ADM2 ID | Mean CO2 |
|---:|---|---|---:|
| 1 | China | CHN.10.9_1 | 199 632 759 |
| 2 | China | CHN.24.1_1 | 141 872 754 |
| 3 | China | CHN.15.7_1 | 133 343 969 |
| 4 | China | CHN.3.1_1 | 120 020 252 |
| 5 | South Africa | ZAF.6.3_1 | 115 805 012 |
| 6 | China | CHN.15.10_1 | 113 811 236 |
| 7 | China | CHN.23.13_1 | 104 862 150 |
| 8 | China | CHN.27.1_1 | 103 411 029 |
| 9 | China | CHN.10.4_1 | 99 511 528 |
| 10 | China | CHN.23.17_1 | 98 720 889 |
| 11 | China | CHN.19.7_1 | 96 086 514 |
| 12 | China | CHN.15.4_1 | 87 744 994 |
| 13 | China | CHN.23.16_1 | 83 558 272 |
| 14 | China | CHN.12.16_1 | 78 899 516 |
| 15 | China | CHN.28.5_1 | 76 522 584 |
| 16 | China | CHN.23.9_1 | 74 562 077 |
| 17 | China | CHN.13.12_1 | 72 518 799 |
| 18 | China | CHN.18.1_1 | 72 069 729 |
| 19 | China | CHN.13.4_1 | 70 898 769 |
| 20 | China | CHN.10.8_1 | 70 602 072 |

*Note:*
CO2 levels are ADM2 means over the panel. Units follow the EDGAR source used for aggregation. Countries are matched from ISO3 codes; unmatched codes remain as ISO3.

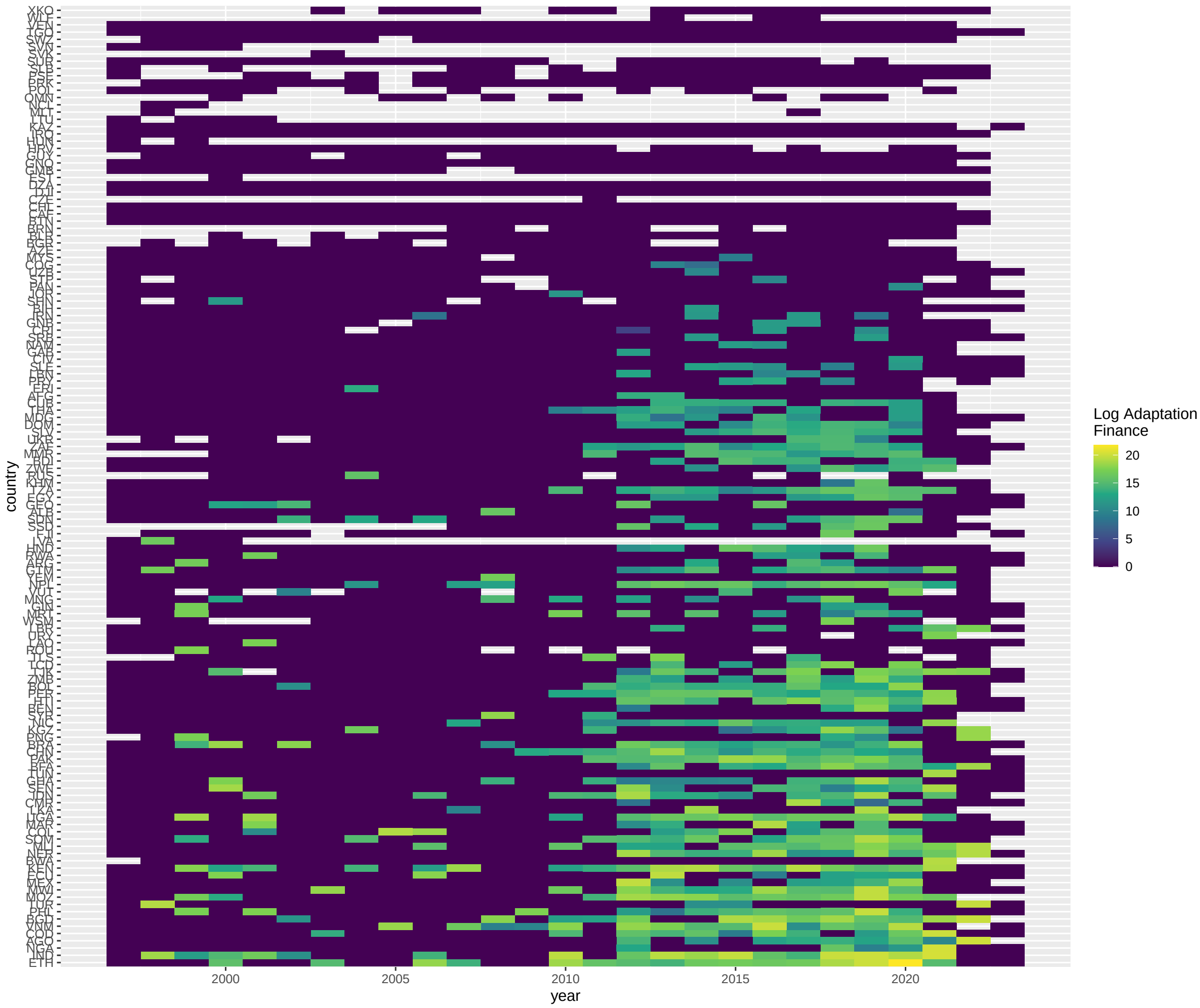


Figure A1: Heatmaps of Adaptation Finance per Country

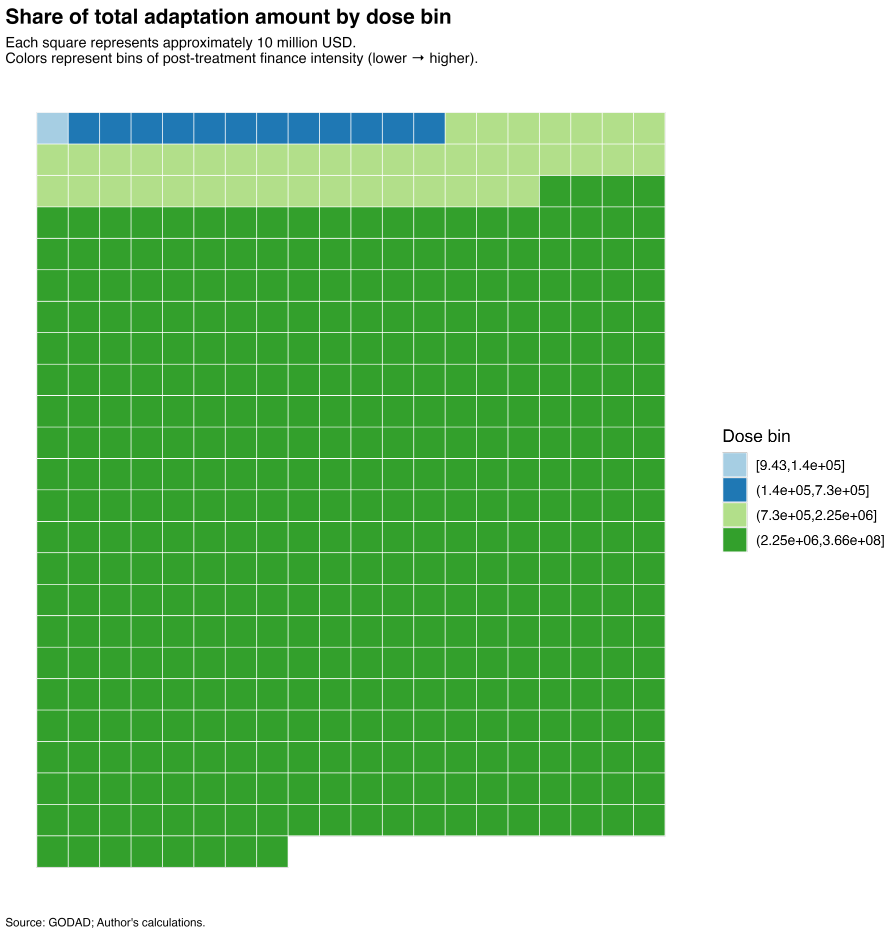


Figure A2: Adaptation bins post treatment intensity repartition

# Reproducibility

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Running under: macOS 26.1

Matrix products: default
BLAS:   /Library/Frameworks/R.framework/Versions/4.4-arm64/Resources/lib/libRblas.0.dyli
LAPACK: /Library/Frameworks/R.framework/Versions/4.4-arm64/Resources/lib/libRlapack.dyli

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time zone: Europe/Paris
tzcode source: internal

attached base packages:
```

```
[1] stats     graphics  grDevices utils     datasets  methods   base

other attached packages:
 [1] forcats_1.0.0     countrycode_1.6.0 patchwork_1.3.0   tibble_3.3.0
 [5] kableExtra_1.4.0  knitr_1.50        fixest_0.13.2     did_2.1.2
 [9] scales_1.4.0      ggplot2_3.5.2     janitor_2.2.1     stringr_1.5.1
[13] sf_1.0-20         arrow_21.0.0.1    readr_2.1.5       tidyr_1.3.1
[17] dplyr_1.1.4       data.table_1.17.0

loaded via a namespace (and not attached):
  [1] DBI_1.2.3            pROC_1.18.5          DRDID_1.2.0
  [4] sandwich_3.1-1       rlang_1.1.6          magrittr_2.0.3
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